

Gaze Capture based Considerate Behaviour Control of Virtual Guiding Agent

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ABSTRACT

Agents in VR have wide application like guidance. Most current agents are passive, so that people should suspend their current tasks and request agents with explicit demand. It is necessary to make agent more actively open the interaction naturally but without being bothering. We propose a virtual guidance agent which provide voice explanation in appropriate timing, using gaze tracking, attention amount estimation and attention driven state machine. We used time-decayed moving average of angle between gaze direction and face front direction. We implemented the method in VR and evaluated effectiveness in virtual guiding tour experimentally.

Index Terms: Human-centered computing—Visualization—Visualization techniques—Treemaps; Human-centered computing—Visualization—Visualization design and evaluation methods

1 INTRODUCTION

Virtual agents appear in our life more and more frequently, and abilities of agents to recognize people's needs or intentions from people's nonverbal behavior and to respond accordingly are becoming more important. In many cases of current human-agent interaction, agents are more likely to be in a passive state, so that people have to take the initiative to ask for help. Therefore, it is necessary for agents to be proactive but not being disturbing at the same time.

To let agents take the initiative, we focus on humans' eye gaze as an implicit expression of a person. We propose a model that enables the agent to react according to player's gaze, by estimating people's visual attention amount, find out the attention target, and explicit it in a threshold-driven state machine.

Our research goal is to propose guidance agent which makes player to feel the agent react in suitable timing. Also, the agent should assist players base on what they are currently interested in and has enough confidence that they need help from players.

2 RELATED WORKS

Bee et al. [1] used of heat map and analysis of gaze fixation can show the social presence experience of a user with the comparison between natural gaze shifts and unnatural gaze. [2] propose the wizard-of-oz approach to analyze when and how to encounter visitors base on their behavior of gaze, trajectory and body direction. Besides analyze by observation, another common way is to use machine learning. The intention is classified as cognitive, affective, and positive by using SVM and individual repeated measures data [4]. However, these researches place more emphasis on the finding features instead of implementing them on human-agent interaction.

Inference of visual attention is often utilized in agent cooperation to assist people. Huang et al.'s [3] used SVM-based model to predict the intention of customers in a sandwich cooperate-making process.

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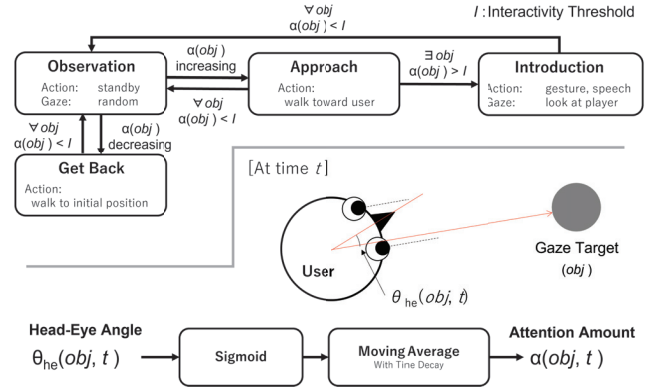


Figure 1: System Overview

Even though, There is still little research aiming at targeting people's attention to be utilized on human-agent interaction.

3 PROPOSAL

We propose a virtual guidance agent which provide voice explanation in appropriate timing, using gaze tracking and attention amount estimation. Figure 1 shows the overview.

As the index of attention level, we use angle between gaze direction and face front direction. To avoid too much effect of eye gaze at each moment, we take moving average for few second. Finally, state machine driven by attention amount determines agent's action.

3.1 Visual Attention Amount Index

The system capture player's head direction, gaze direction, object that player looking at. While the player is looking particular object obj , angle between head and eye direction $\theta_{he}(t)$ is considered to show how seriously the player is looking the object. Therefore, we employ this angle as the attention index. The head-eye angle value transformed by sigmoid function is calculated as equation (1). Then, moving average over time is applied to compute attention amount index $\alpha(obj, t)$ (equation (2)). Object which has maximum attention amount becomes the target of explanation.

$$S(obj, t) = \begin{cases} 1 - \left(\frac{2}{1 + e^{-g\theta_{he}(t)}} - 1 \right) & \text{if user is looking } obj \\ 0 & \text{else} \end{cases} \quad (1)$$

$$\alpha(obj, t) = \sum_{i=0}^N 0.2t_{w_{N-i}} S(obj, t) \quad (2)$$

3.2 State Machine Driven by Attention Amount

Figure 1 shows the state machine. Agent's behaviors are:

- (1) **Observation** The agent look around the user and surroundings.
- (2) **Approach** The agent approaches the user to show willingness of assistance and participation.

Table 1: Questionnaire

assistance	
Q1	I think I get assistance from guide.
Q2	The guide can react to my interest precisely.
bother	
Q3	I feel the guide response in time.
Q4	I feel the guide response too late.
Q5	I feel the guide response too quick.
Q6	I feel bothered by guide.
Q7	I feel tired after the visit.
naturalness	
Q8	I think the guide behaves naturally.

(3) **Introduction** The agent talk to the user for guidance.

(4) **Get Back** The agent gets back to a certain standby position.

The transitions of states are triggered by attention amount and threshold. The status transit from Observation to Approach is triggered when the total attention amount is increasing and above a certain amount. The transition from Approach to Introduction happens when the attention target surpass interactivity threshold. The transition from Observation to Get Back starts when the total attention amount is decreasing and under a certain amount. The transition (4) occurs when no attention amount exceeds the threshold.

To adjust proper timing to respond according to situation, we use a dynamic threshold along time which we call interactivity threshold(I) as equation (3). C_{gaze} is total gaze count that player looked specific object during the time window.

$$I = \begin{cases} (B + W_c C_{gaze}) D_{gaze} & \text{if player is looking agent} \\ B + W_c C_{gaze} & \text{else} \end{cases} \quad (3)$$

In equation (1)(2)(3), g , N , D_{gaze} , B , W_c are parameters which we determine from data. We recorded human-human guidance situation in VR, while one did agent's role and other did user's. Then we optimized parameters so that the result reproduce recorded data the most. Finally, we set $g = 0.043$, $N = 26$, $B = 68.56$, $W_c = 0.79$.

4 EVALUATION

We evaluated our model by user test comparing it with two other models. Impression of the agents are judged by subjects in 3 aspects: assistance, not bothering and naturalness. The interaction and gaze capture in VR are done with HTC VIVE Pro Eye.

We compared our method with two other methods:

PROPOSAL proposed method

SIMPLE do explanation when user looking object for fixed time

WOZ human('wizard') determine timing and trigger agent's speech manually while observing player's action

The participants are asked to visit each experiment for around five minutes. Then the participants are asked to fill out the questionnaire shown in Table 1 respectively in the three experiments. We ask participants 8 questions using 7-point Likert scale and 2 questions for user feedback. There are 10 subjects participating the experiment, and they are able to understand daily conversation in English. The subjects are aged between 10-40 (5 female, 5 male).

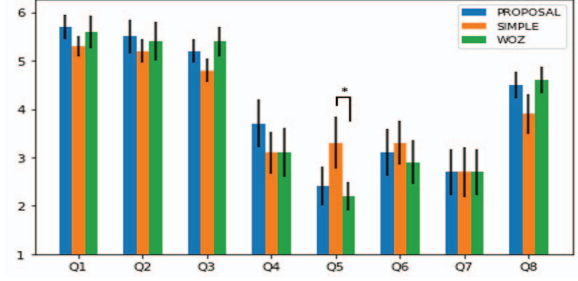


Figure 2: Questionnaire Result (mean and standard error of score)

4.1 Result and Discussion

Figure 2 shows the result. Most people somewhat agree or agree more with the assistance of guidance. The final score of PROPOSAL(5.7) is similar to WOZ(5.6) and seems to outperform SIMPLE(5.3) in Q1. By t-test for comparing between groups($p < 0.05$), SIMPLE responded significantly quicker than WOZ, while PROPOSAL didn't. Participants also seem to regard PROPOSAL and WOZ to be more in time than SIMPLE. This suggest that SIMPLE may respond too quick and be more bothering than the other two.

Based on the participant's feedback, proposed model has some advantages than SIMPLE. The introduction of agent continues longer and agent starts her speech while looking at her. On the other hand, SIMPLE model explains shorter and has pause between speech, resulting from the interrupt of continuous gaze. Furthermore, some participants feel disturbance with the speech since it starts responding right after direct gaze at the object. However, our proposed model sometimes makes participants feel the lateness of reaction because of longer perception of the agent. This disadvantage also occurs in WOZ but with less lateness.

5 CONCLUSION

We proposed a novel model for agents to assist player on their initiative by estimation of player's attention from real-time gaze tracking. It allows players to experience guiding tours without direct input toward agents which is often necessary in human-agent interaction. As an attention amount index, we utilized time-decayed moving average of angle between gaze direction and face front direction. Evaluation result shows that participants can get assistance in the experience and somewhat agree it is not bothering than straightforward method.

In our current model, parameter sets are fixed. It is not customized for players and not all players have the preference using our calculation of threshold. As the next step, it is necessary to design a way to simply adjust the values for the individual difference which may possible be learned by agent before or during the interacting phase.

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